Unit → 2

Date / /
Page No. 40

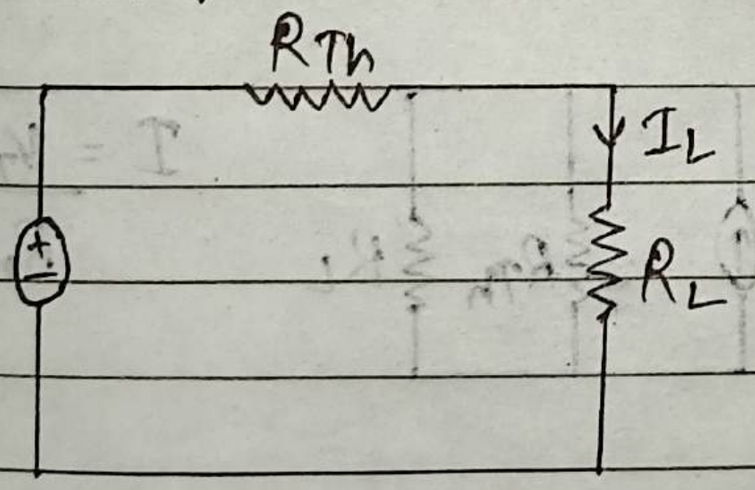
* Network Theorems →

Thevenin's Theorem
Norton's Theorem
Maximum power Transfer Theorem
Superposition Theorem

Thevenin's Theorem →

Thevenin's theorem states that linear bilatral network can be replaced by an equivalent circuit consisting of voltage source (V_{TH}) in series with resistance (R_{TH}), V_{Th} is an open circuit voltage b/w the terminals when the independent source are turned off.

This theorem can only be applied to linear bilatral network.

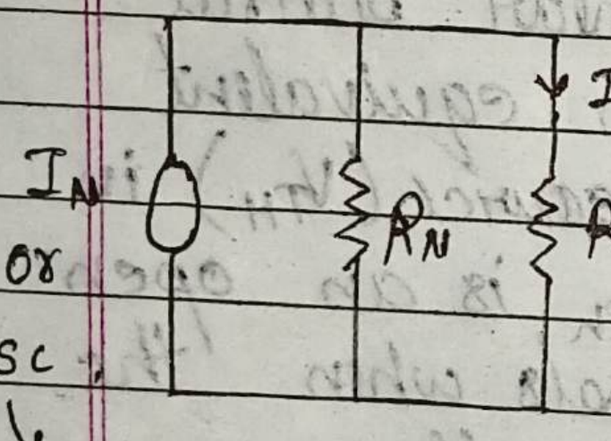


$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

Date / /
Page No. 41

→ Norton's Theorem ←

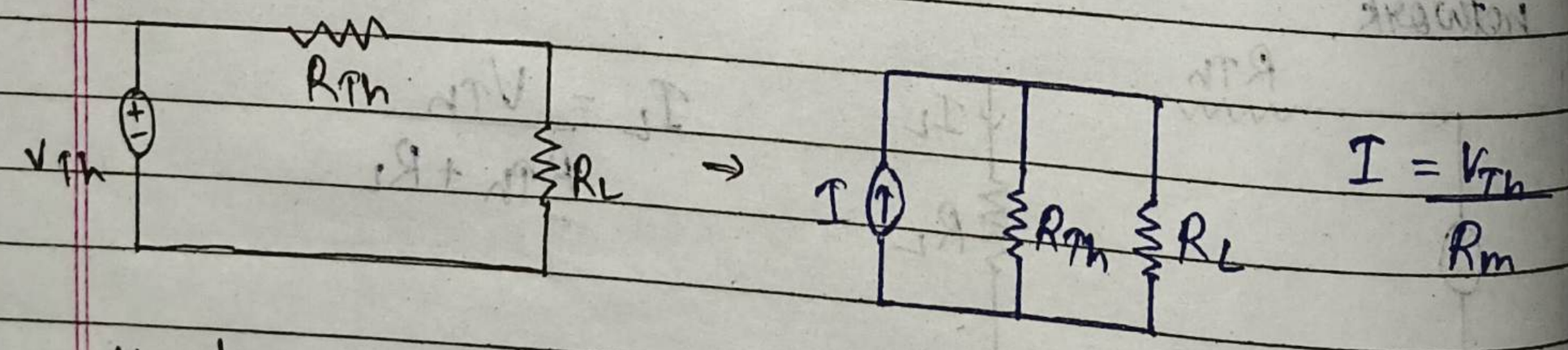
Norton's theorem states that any linear bilateral network can be replaced by an equivalent circuit consisting of current source (I_N) in parallel with (R_N) is a short circuit current between terminals and R_N is an equivalent resistance b/w the terminals when the independent source are turned off.



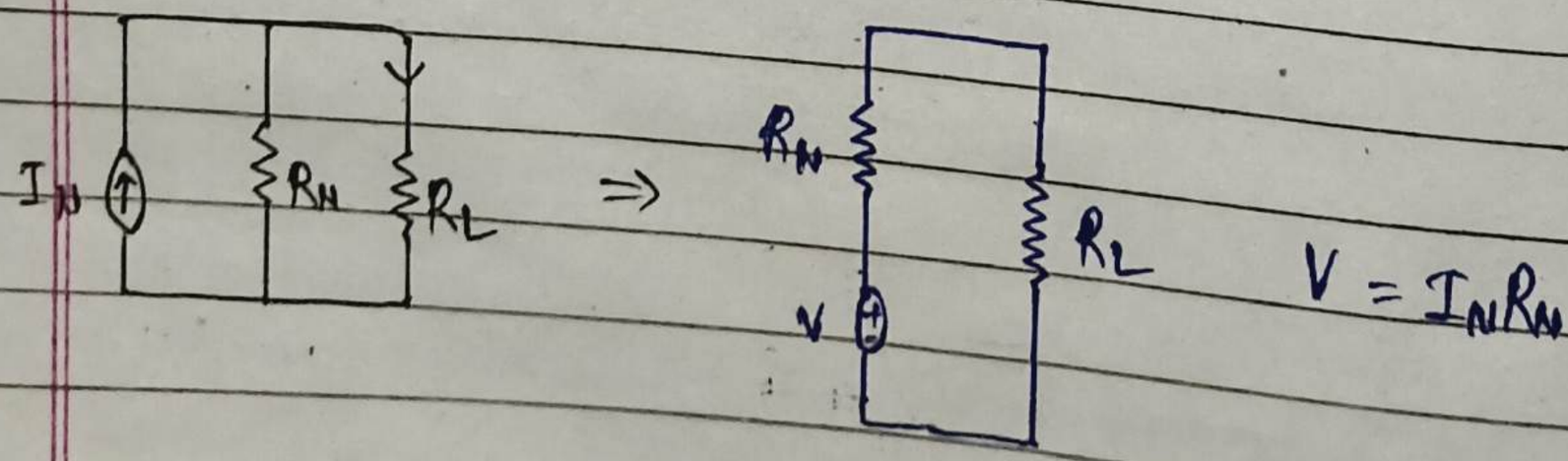
$$I_L = I_N \times \frac{R_N}{R_N + R_L}$$

↳ Short circuit current

Thevenin to Norton →

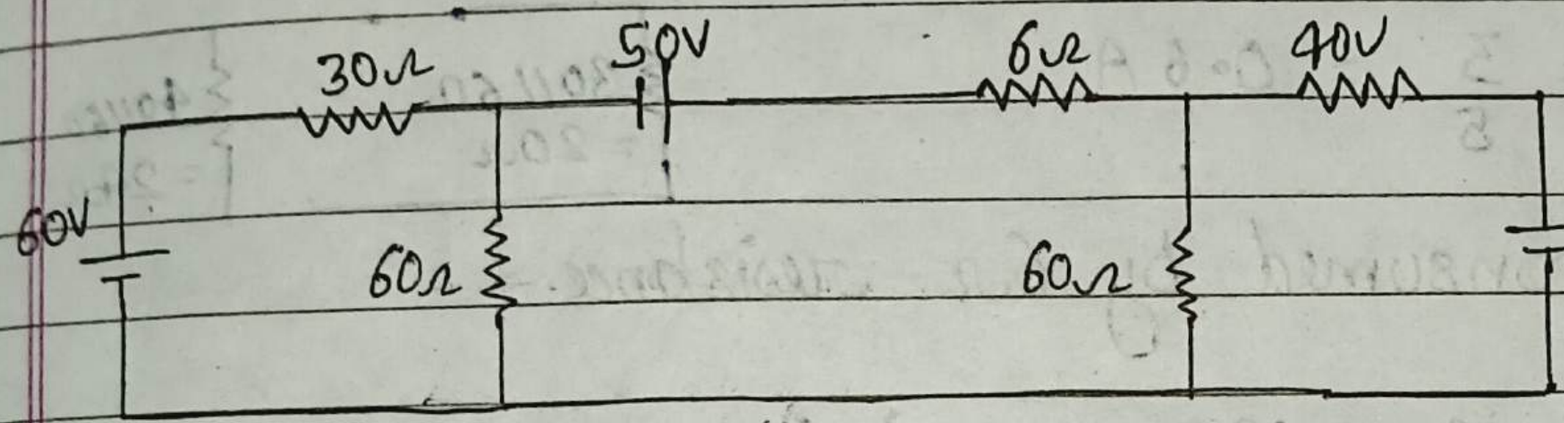


Norton to Thevenin →



↳ Thevenin and Norton's circuit

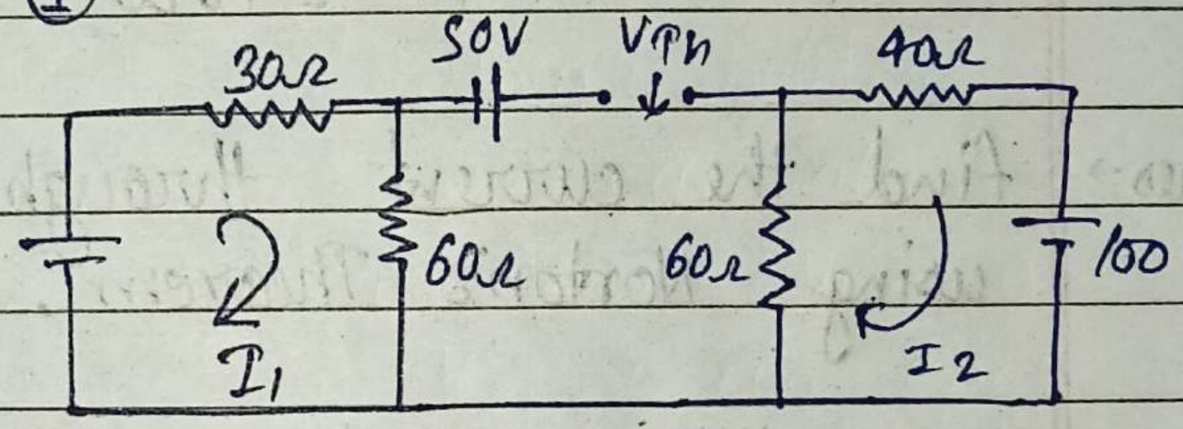
Ques. Find the current through 6Ω resistance and power consumed by 6Ω resistance, using ? thevenin?



Applying KVL in mesh (1)

$60 - 90I_1 = 0$

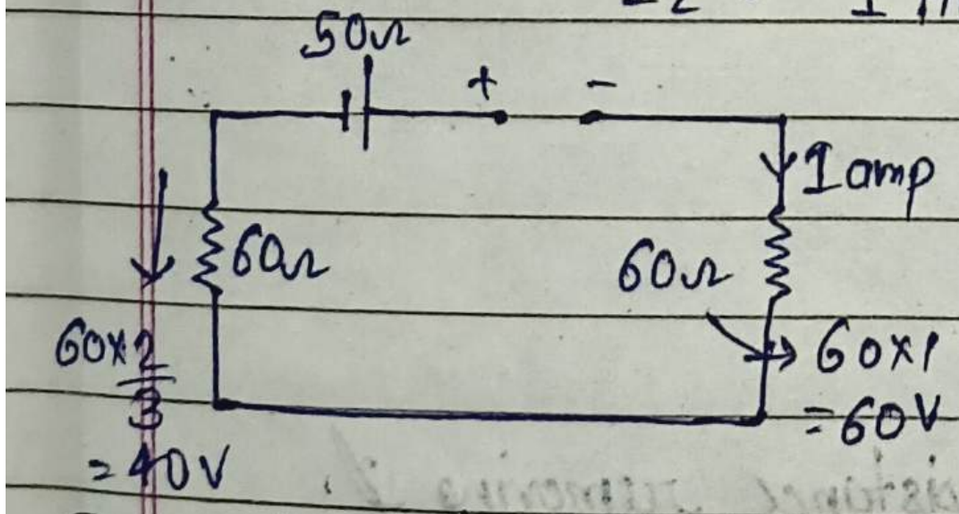
$I_1 = \frac{2}{3}$ Amp



K.V.L in mesh (2) →

$-100 - 100I_2 = 0$

$I_2 = -1$ Amp

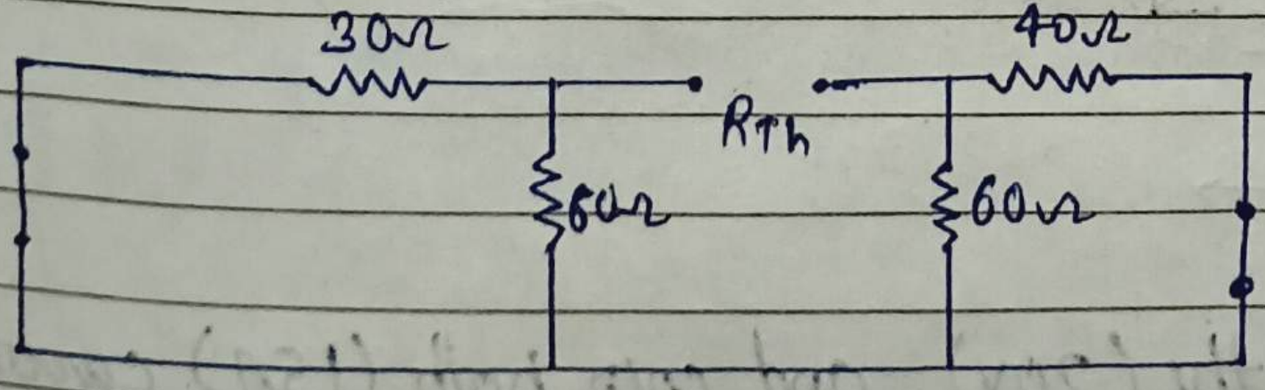


Now in middle mesh

$-V_{Th} - 60 + 40 + 50 = 0$

$V_{Th} = 30V$ .....

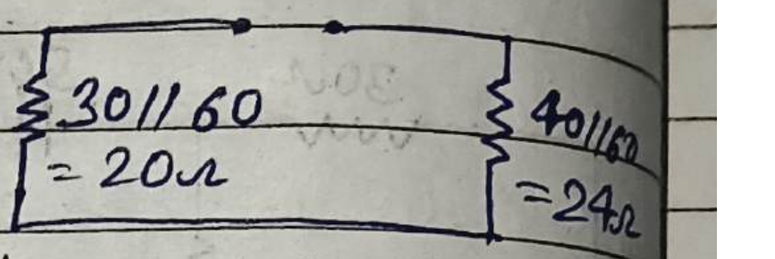
Calculation of R_{Th} → Replace will idea voltage source



$R_{Th} = 20 + 24 = 44\Omega$

$$I_L = 6\Omega = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{44+6} = \frac{30}{50}$$

$$I_L = \frac{3}{5} = 0.6\ A$$

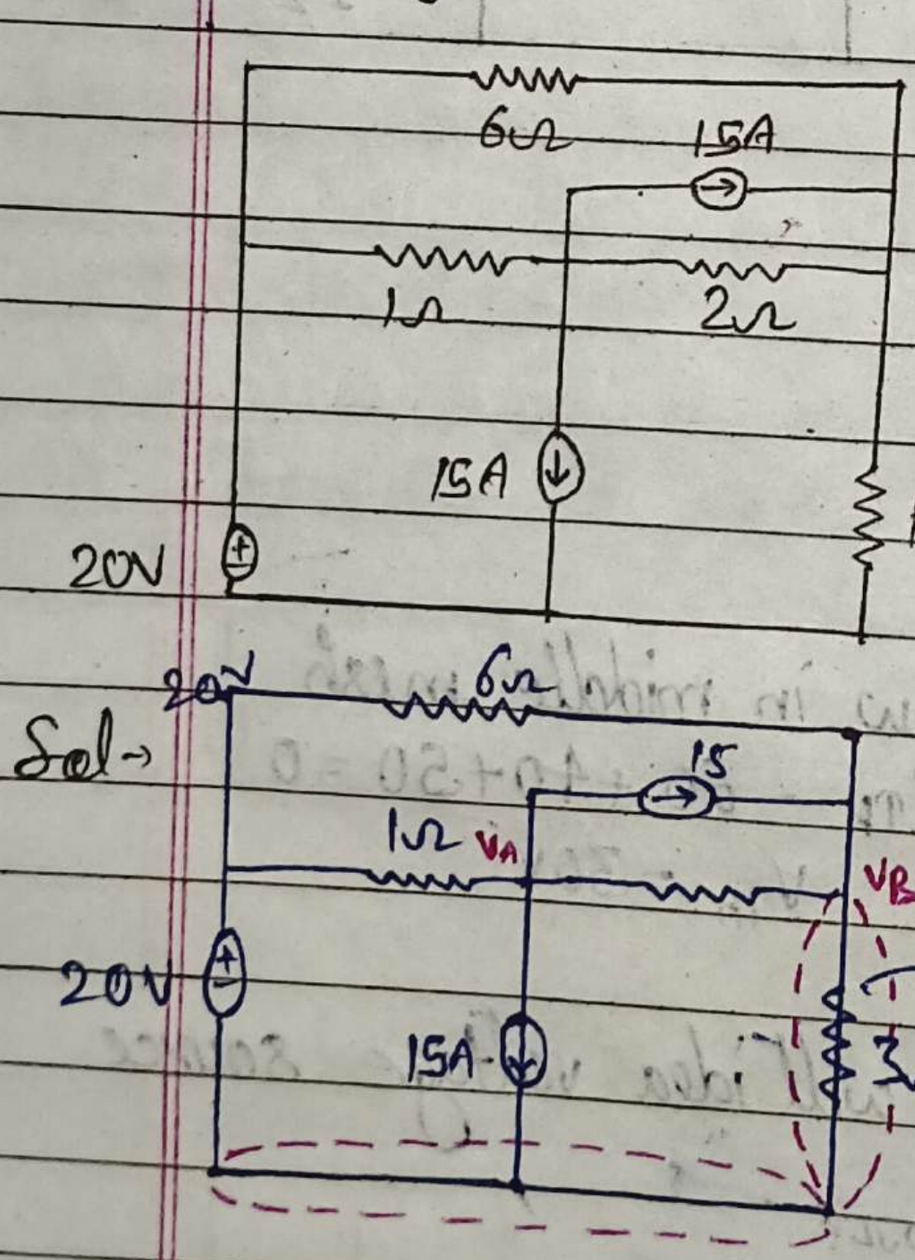


Power consumed by 6Ω resistance →

$$P = i^2R = (0.6)^2 \times 6$$

$$P = 2.16\ W$$

Ques→ find the current through load resistance by using Norton's Theorem.



Sol→

(#) calculate R_N →

Short-circuit the (20V) and open both (15A) current source →

$R_{series} = 1+3 = 3\Omega$

$R_N = 6||3 = \frac{6\times 3}{6+3} = 2$

$R_N = 2\Omega$

Calculate (V_{oc}) open circuit voltage →

KCL at node (A)

$$\frac{V_A - 20}{1} + \frac{V_A - V_B}{2} + 15 + 15 = 0$$

$$3V_A - V_B = -20 \longrightarrow (1)$$

KCL at node (B)

$$\frac{V_B - 20}{6} + \frac{V_B - V_A}{2} - 15 = 0$$

$$-3V_A + 4V_B = 110 \longrightarrow (2)$$

Put $V_B = 3V_A + 20$ in eqn (2)

$$-3V_A + 4(3V_A + 20) = 110$$
$$-3V_A + 12V_A + 80 = 110$$
$$9V_A = 30$$
$$\left[V_A = \frac{10}{3}\ V\right]$$

$$\left[V_{OC} = V_B = 3\left(\frac{10}{3}\right) + 20 = 30V\right]$$

Calculate of I_N →

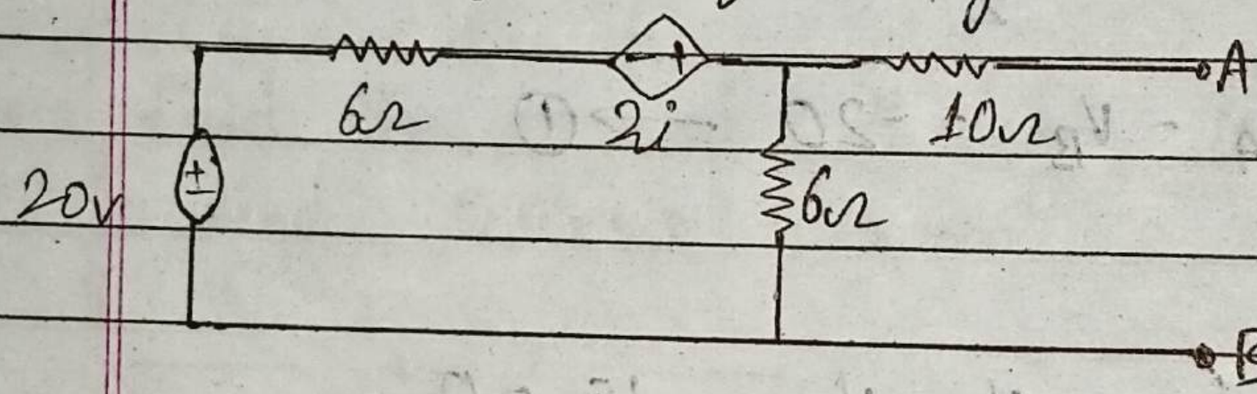
$$\left[I_N = \frac{V_{OC}}{R_N} = \frac{30}{2} = 15\ Amps\right]$$

* Calculate of I_L →

$$I_L = \frac{I_N \times R_N}{R_N + R_L}$$

$$I_L = \frac{15 \times 2}{2+3} = \frac{15 \times 2}{5} = 6 \text{ Amp}$$

Ques Determine the therenin circuit and norton circuit of the following circuit?



Calculation of V_{Th}

Applying KVL in Loop →

$$20 - 6i + 2i - 6i = 0$$

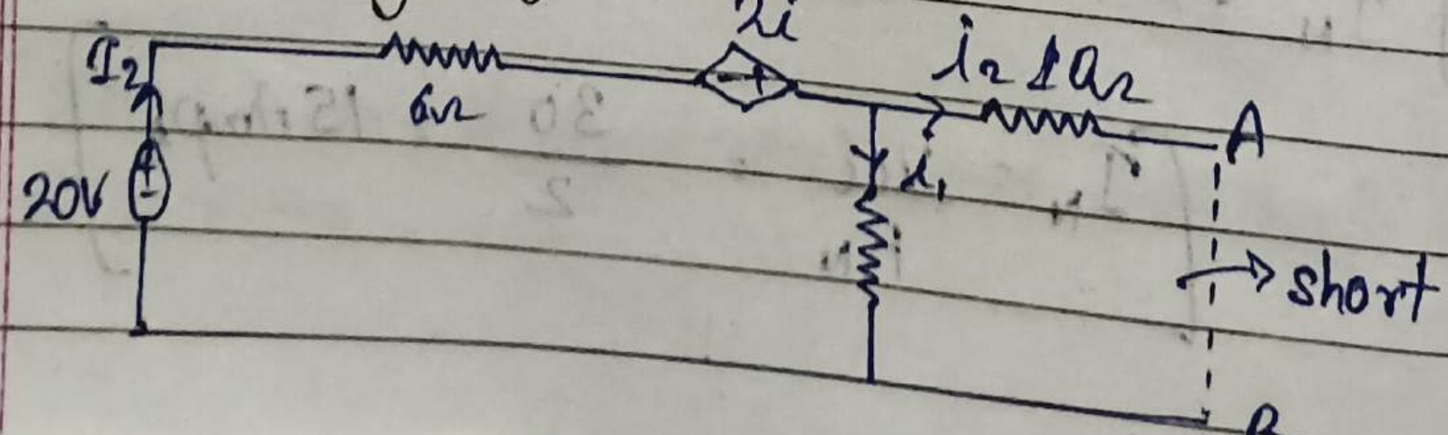
$$i = 2A$$

voltage at the top of the vertical 6Ω resistor →

$$V = 6i = 6 \times 2 = 12V$$

$$[V_{Th} = 12V]$$

Calculating of I_{SC}



$$i_1 = \frac{V}{6} \qquad i_2 = \frac{V}{10}$$

current left through hand (6Ω) resistor is the sum of the currents through the vertical 6Ω and horizontal 10Ω resistor.

$$I_L = \frac{V}{6} + \frac{V}{10}$$

$$I_L = \frac{5V + 3V}{30} \Rightarrow \left[I_L = \frac{4V}{15} \right]$$

Applying in KVL in circuit

$$V = 20 - 6I_2 + 2i$$

$$V = 20 - 6\left(\frac{4V}{15}\right) + 2\left(\frac{V}{6}\right)$$

$$V = 20 - \frac{8V}{5} + \frac{V}{3} \Rightarrow V + \frac{19V}{15} = 20$$

$$\frac{34V}{15} = 20 \Rightarrow \left[V = \frac{150}{17} = 8.824\ V \right]$$

→ calculating of I_{sc} →

$$I_{sc} = \frac{V}{10} = I_{sc} = \frac{150/17}{10} = \frac{15}{17} = 0.8824\ Amp$$

→ calculating of R_{Th} →

$$R_{Th} = \frac{V_{Th}}{I_{sc}} = \frac{12}{15/17}$$

$$\left[R_{Th} = \frac{12 \times 17}{15} = 13.6\ \Omega \right]$$

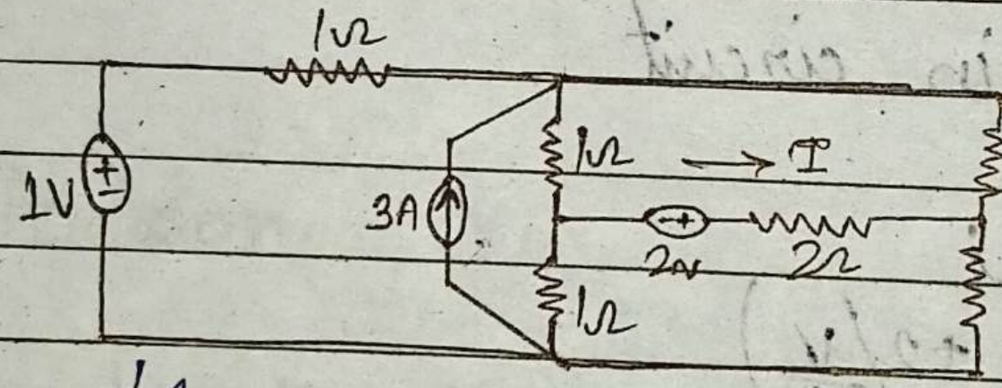
Date / /
Page No. 47

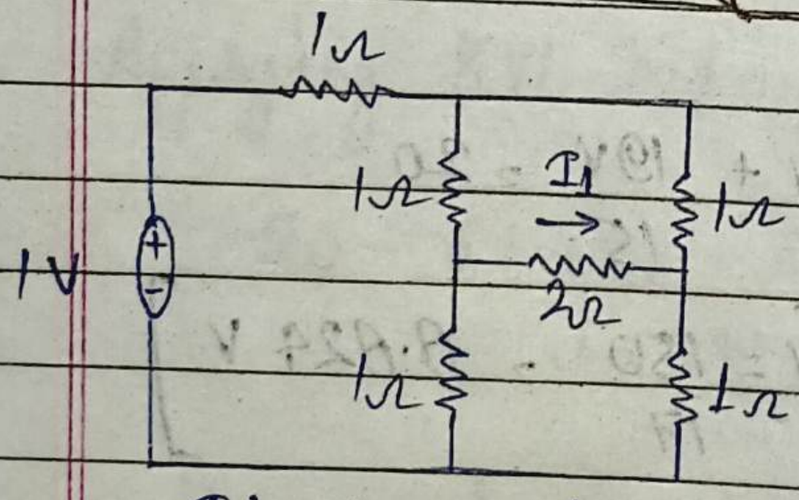
→ Superposition Theorem ←

→ It states that the voltage across an element in a linear circuit is algebric sum of voltage across or circuit due to each independent source acting alone.

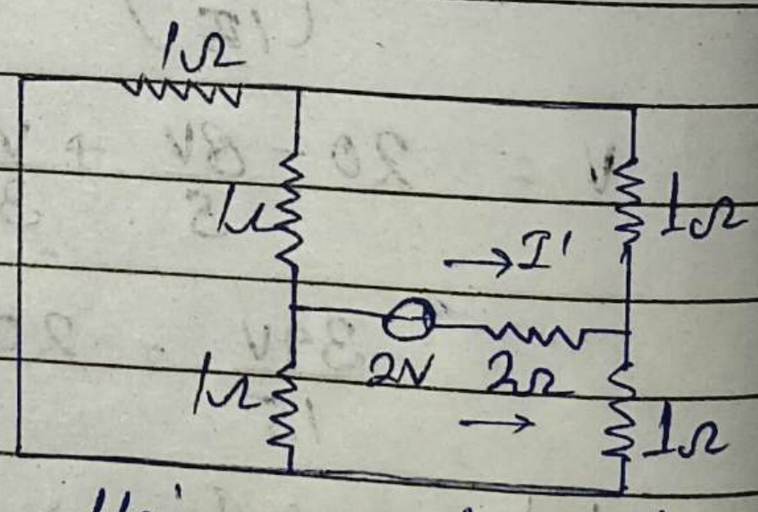
Eliminate independent voltage source by short circuit and independent current source by open circuit.

find → current 'I' in the following circuit →





This is condition of wheatstone brid



Using mesh analysis

$$Req = 1+(1+1)\|(1+1)$$

$$1+(2\|2)$$

$$1+\frac{2\times2}{2+2} = 2$$

$$[Req = 2\Omega]$$

due to wheat stone bridge con^ $[I' = 0]$

$$+2 - 1I'' - 1I'' = 0$$

$$2 = 2I''$$

$$[I'' = 1\ amp]$$

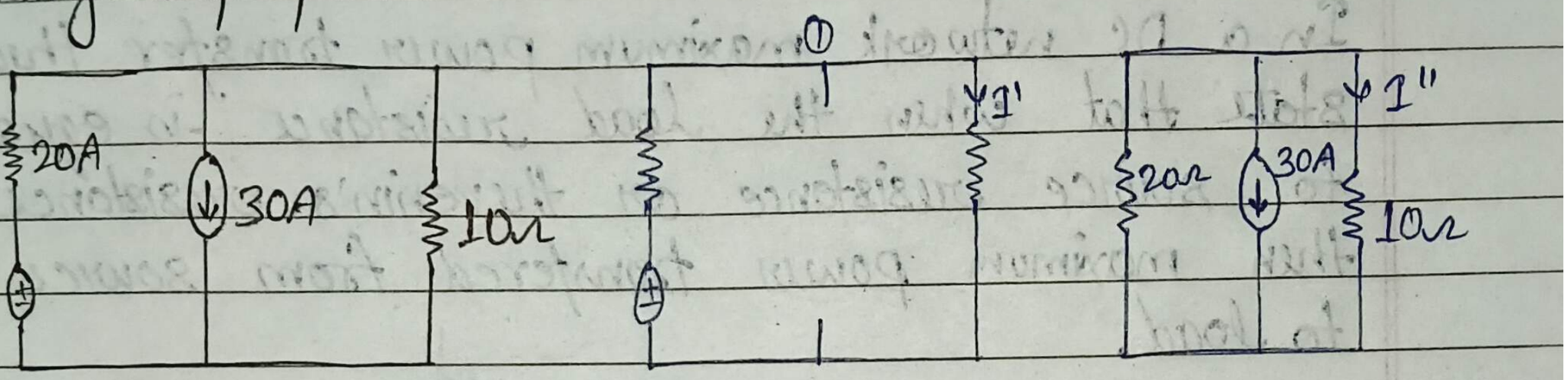
Required current →

$$I = 0+1$$

$$I = 1\ amp$$

Date / /
Page No 40

Find the current flowing through 10Ω resistance using superposition Theorem.



Current across 10Ω

$I' = \frac{60}{30} = 2\text{Amp}$

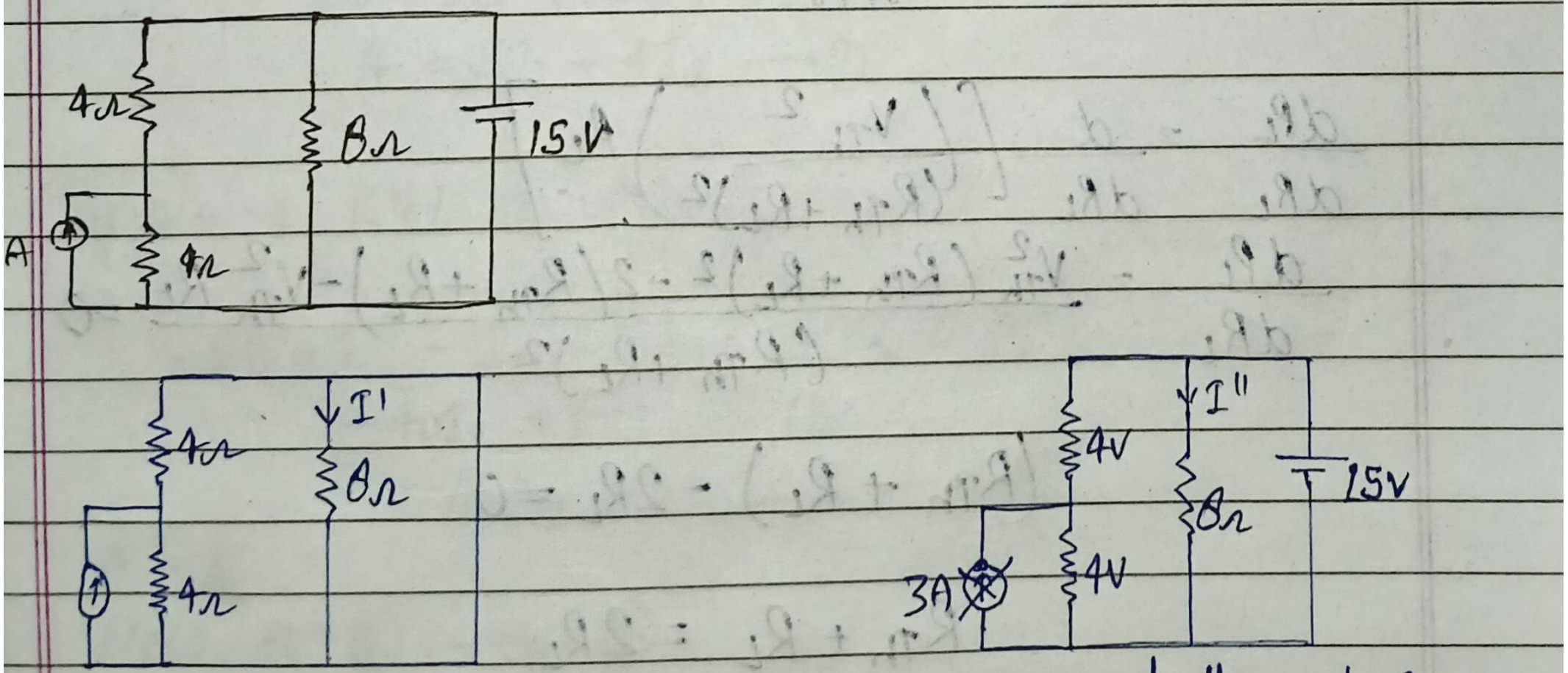
$\boxed{I' = 2}$

$I'' = -30 \times \frac{20}{20+10}$

$\boxed{I'' = -20\text{Amp}}$

$\boxed{I_{10\Omega} = I' + I'' = 2 - 20 = -18\text{A}}$

2→ find current in 8Ω using Superposition theorem.



Current follow less resistance path

$\boxed{I'_{8\Omega} = 0\text{ Amp}}$

Current through 8Ω

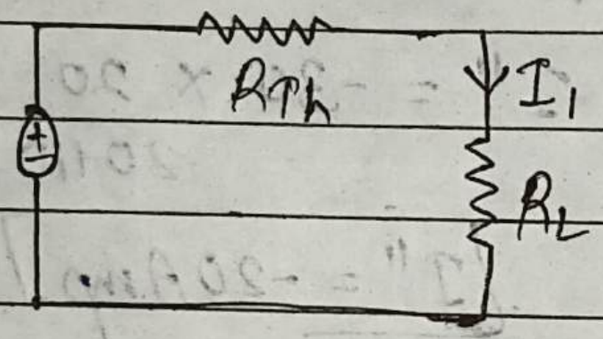
$I_{8\Omega} = I' + I''$

$= 0 + 15/8$

$\boxed{I_8 = \frac{15\text{A}}{8}}$

→ Maximum Power Transfer Theorem ←

In a DC network maximum power transfer theorem state that when the load resistance is equal to source resistance or thevenin's resistance then maximum power transfered from source to load.



$$I_1 = \frac{V_{Th}}{R_{Th} + R_L}$$

$$P = i^2 R$$

$$P_L = i_L^2 \times R_L$$

for maxima $\frac{dP_L}{dR_L} = 0$

$$\frac{dP_L}{dR_L} = \frac{d}{dR_L}\left[\left(\frac{V_{Th}^2}{(R_{Th}+R_L)^2}\right) R_L\right]$$

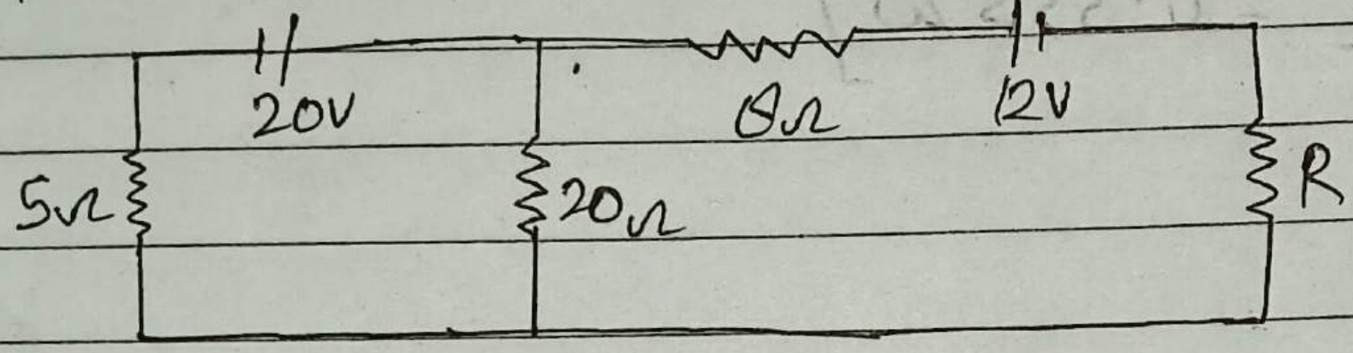
$$\frac{dP_L}{dR_L} = \frac{V_{Th}^2 (R_{Th}+R_L)^2 - 2(R_{Th}+R_L) - V_{Th}^2 R_L = 0}{(R_{Th}+R_L)^2}$$

$$(R_{Th} + R_L) - 2R_L = 0$$

$$R_{Th} + R_L = 2R_L$$

$$\boxed{R_L = R_{Th}}$$

eus Determine the value of R such that maximam power value transfer from it.



$R_{Th} = R = R_{eq} = (5 \| 20) + 8$

$= \frac{5 \times 20}{25} + 8 = 4 + 8 = 12\Omega$

$R_L = 12\Omega \quad P_L = (I_L)^2 \times R$

applying KVL in loop ①

$20 - 20(I_1 - I_2) - 5I_1 = 0$

$20 - 25 + 20I_2 = 0$

$4 = 5I_1 - 4I_2 \rightarrow ①$

applying K.V.L in loop ②

$-20(+I_2 - I_1) - 8I_2 - 12I_2 = 12$

$-40I_2 + 20I_1 = 12$

$-10I_2 + 5I_1 = 3 \rightarrow ②$

Now, eqn ② − eqn ①

$5I_1 - 4I_2 = 4$

$\underline{-10I_2 + 5I_1 = 3}$

$6I_2 = 1$

$\left[I_2 = \frac{1}{6} \text{ Amp}\right]$

$$P_{R_L} = (I_L)^2 \times R = \left(\frac{1}{6}\right)^2 \times 12 = \frac{1}{3}$$

$$[P_L = 0.333\ W]$$